

SciOdyssey.

A research layer over your agent harness.

PRESERVE THE RESEARCH WORLD · RESET THE RESEARCHER

ROBUST

SUPERVISABLE

BROAD SEARCH

SAME EVIDENCE
FRESH JUDGMENT
A WIDER SEARCH

Research
Runs
Longer.

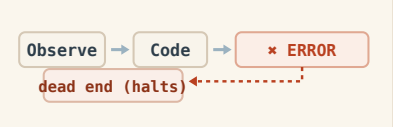
01 / PROBLEM

Why current agents fail.

Sustained autonomous research breaks in three different ways.

01

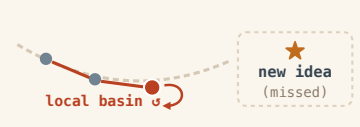
Closed-world fragility
Fixed workflows shatter on unexpected runtime errors.



Unhandled runtime faults halt unassisted runs

02

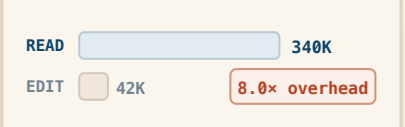
Trajectory momentum
A single context carries past momentum into local basins.



22 runs trapped tweaking parameters in same basin

03

Context inflation cost
Reconstructing context costs far more than the edit itself.



10 files read to change 1 line; burns token budget

DESIGN RESPONSE Keep the world. Reset the trajectory. · Decouple persistent memory from reasoning

- Auto-heal runtime faults
- Fresh reasoning each cycle
- Reusable disk state (no token waste)

02 / SYSTEM

One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

DUAL-LIFETIME LOOP
Persistent world · ephemeral trajectory · audited write-back

Fresh Scientist
Owns scientific judgment inside one trajectory: hypotheses, code, experiments, interpretation, and pivots.

META Review
Audits claims against artifacts, scopes conclusions, and decides what is durable enough to survive.

SELECTIVE PERSISTENCE
Only audited empirical findings, code state, and bounded priors write back to the shared world.

PERSISTENT RESEARCH WORLD
What survives across trajectories

A · EVIDENCE
Knowledge + Ledger
Metrics, failures, reports, and experiment receipts in `ledger.jsonl`.

B · PRIORS
Brief + Hypotheses
Current understanding and exploration leads in `brief.md`, open to revision.

C · STATE
Code + State
Retained implementation, serialized model weights, checks, and git provenance.

CONTEXT RESET
The next Scientist reads verified experience from disk, but does not inherit the previous reasoning trace.

INPUTS
TASK
DATA
EVALUATOR
BUDGET

WORKTREE ISOLATION
Independent git worktrees isolate branches and preserve reproducible artifacts.

SELF-HEALING
Traceback → patch → rerun keeps autonomous research moving after runtime faults.

PARALLEL SYNTHESIS
Completed branches can be compared post-hoc; useful ideas survive even when a branch loses.

RUNTIME SUBSTRATE

Runtime Substrate

The next Scientist inherits **verified experience** — not prior reasoning momentum.

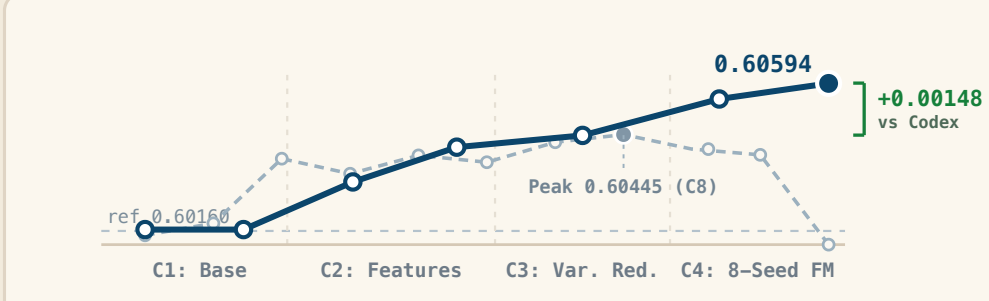
Decoupled Memory · Audited Persistence · Fresh Scientific Judgment

03 / EVALUATION

Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

01 MEASURED RESULT
The model got better.
0.6016000 → **0.6059363**
Official FM baseline SciOdyssey submission
+0.0043363 Primary improvement
GAUC **0.6728421** (+0.0054) nDCG@5 **0.5390304** (+0.0033) LogLoss **0.38102** (-0.0032)
Feature Engine: 46 categorical cross-interactions + log1p transforms
Architecture: 8-seed Factorization Machine ensemble (NumPy / CPU)
100% FULLY AUTONOMOUS 48.24M LLM tokens E001-E013 (4 cycles)
Monotonic gain: 0.6016 → 0.6041 → 0.6059 (100% CPU inference, zero GPUs).

02 SUSTAINED SEARCH
The search stayed productive.
● SciOdyssey ● Codex Goal

4 cycles 7 Full evaluations **E003-E013** frontier **+0.00148** over Codex

03 SELF-HEALING & GATEKEEPING

The agent recovered from failure.

TIER 1 · RUNTIME SELF-HEALING CRASH AUTO-RECOVERY
FAILURE Cycle 1 `numpy.float32` serialization crash
ACTION Scientist parsed traceback, auto-patched scalar helper
OUTCOME 0 human bugfixes · Unattended rerun succeeded

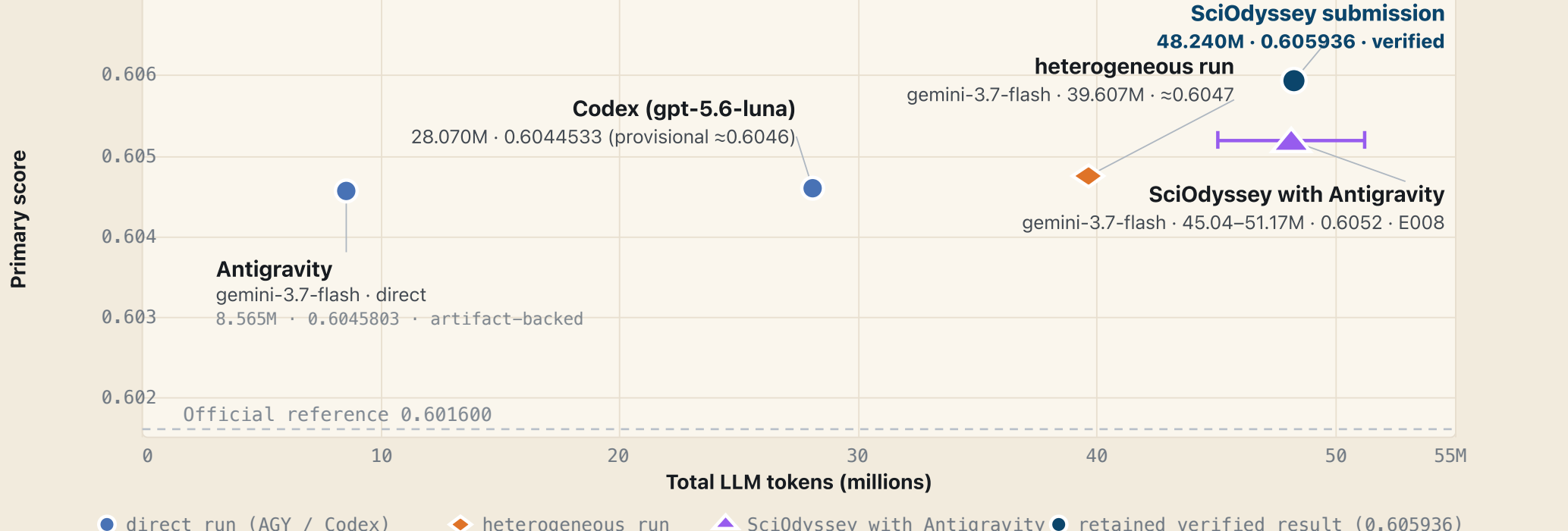
TIER 2 · META GATEKEEPING SPURIOUS STATE REJECTION
FALSE CLAIM Scientist claimed BPR gain on medium proxy
ACTION META audited `system/data.py`, caught `UNK` leak
OUTCOME Promotion blocked · Zero contaminated state

Empirical receipts preserved: runtime self-healing telemetry and independent code-level proxy audits.

05 RESOURCE ACCOUNTING

Score vs token investment.

~\$10 cost 1 × M2 0 GPU-hr 1h 55m
KuaiRand-Pure · Public validation · Inclusive input+output tokens

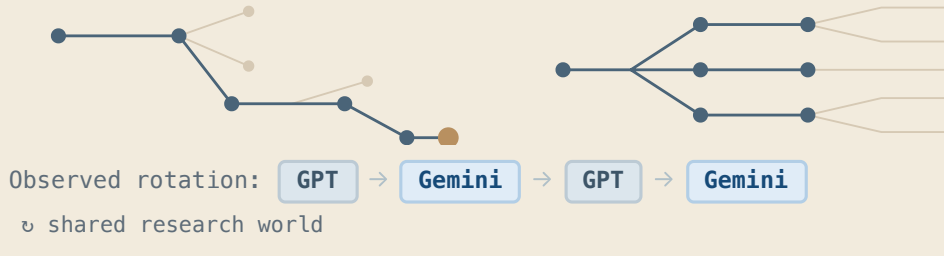


● direct run (AGY / Codex) ◆ heterogeneous run ▲ SciOdyssey with Antigraivty ● retained verified result (0.605936)

All results **verified** on KuaiRand-Pure official validation split with **complete reproducible artifacts**.

04 / FINDINGS

What the run revealed.

01 · MODEL DIVERSITY
Model diversity can help break plateaus.
GPT 5.6 Deepen proven paths Gemini 3.7 Open new mechanisms

Gemini-only **0.6052** → GPT+Gemini **0.6059**

02 · PARALLEL SYNTHESIS
A losing branch can still improve the winner.
A **0.6048** Redundant B **0.6048** Chosen ✓ C **0.6039** Useful ideas
Reviewer Synthesize B+C → Fresh Scientist **0.6055536**
Selection keeps useful knowledge, not just the top score: Branch C contributed pairwise ranking and rank normalization.
Branch B+C Ideas → Synthesis **0.6055536**

05 / EXPERIENCE

Start your journey.

task.md
What to solve
Formal task contract: research objectives, dataset inputs, compute constraints, and official evaluation protocol.

- Dataset paths & split integrity
- Official sandbox evaluator

PERSONAL.md
How to work
User research preferences, model guidance, exploration style, tool permissions, and supervisory checkpoints.

- Model rotation (GPT / Gemini)
- Supervision & budget thresholds

01 Contract task.md → 02 Probe step → 03 Search run → 04 Branch parallel → 05 Inspect gui

AUTONOMOUS RUNTIME
Zero manual intervention · Closed-loop agent discovery
1 Unattended Search
Multi-cycle overnight execution; runtime auto-heals unexpected faults without human intervention.
2 Agentic Supervision
Autonomous META agent independently audits empirical artifacts before durable write-back.
3 Parallel Synthesis
Dispatches isolated worktrees; Reviewer agent synthesizes winning mechanisms across branches.

DECLARATIVE PROTOCOL
Contract-driven autonomy: Set boundary in `task.md`; SciOdyssey autonomously iterates, heals, evaluates, and records evidence.

Autonomous by default.

Supervisable by design.

github.com/zc6600/research-agent-kuairand

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Sources: slides/slides.md · docs/ARCHITECTURE.md · docs/FINAL_REPORT.md · KuaiRand-Pure submission ledger

TEAM GOOD4AI · ROBUST · SUPERVISABLE · BROAD SEARCH

A Persistent Research World for Autonomous Science

PROJECT WIKI & DEMO

sciodyyssey.zc6600.wiki

Scan for live report & artifacts